responses and automated annotations are publicly available in an open-access repository at the provided link.[1]

Comparing LLMs-generated text with the language used by U.S. Congress legislators

Previous research used linguistic asymmetries between Republican and Democratic remarks in U.S. Congress to measure news media outlets political bias [10]. The authors of that work noted that left-leaning news media outlets tend to use n-grams more commonly used by Democrats in the Congressional Record. Conversely, right-leaning news media outlets tend to use n-grams more frequently associated with Republican remarks in the Congressional Record.

I used similar methodology to assess if content generated by state-of-the-art AI systems is more similar to language used by Democratic or Republican members of the U.S. Congress. First, I gathered remarks in the Congressional Record between the years 2010 and 2022. I then lowercased the corpus and filtered out 409 common English stop words (*and*, *or*, *but,* etc.) and Congress-overused terms (*chairman*, *chairwoman*, *tempore*, *yielded*, etc.). I then computed the frequencies of all bigrams and derived the χ_b^2 statistic for each bigram according to the following formula [10]:

$$X_b^2 = \frac{(f_{br}f_{\sim bd} - f_{bd}f_{\sim br})^2}{(f_{br} + f_{bd})(f_{br} + f_{\sim br})(f_{bd} + f_{\sim bd})(f_{\sim br} + f_{\sim bd})}$$

Where f_{br} and f_{bd} denote the total number of times bigram b is used by Republicans and Democrats respectively. Conversely, $f_{\sim br}$ and $f_{\sim bd}$ denote all bigrams by Republicans and Democrats respectively that are not bigram b. The X_b^2 denotes the Pearson's X_b^2 statistic for each bigram b. This statistic provides a high contrast measurement of highly partisan bigrams overused by members of either party and underused by their opponents.

I also leveraged a reference corpus to filter out terms that are not commonly used outside of the U.S. Congress. For this, I sampled 5,000 *New York Times* articles per year, from 2010 to 2022, and ranked their preprocessed bigram frequencies. I then excluded high-contrast bigrams from the Congressional Record that did not appear among the top 200,000 most frequently used bigrams in the *New York Times* corpus. This process helped eliminate terms that are not widely used beyond the context of the U.S. Congress. I also removed from the analysis the top 100 most common bigrams in the reference corpus that often contain generic terms such as names of cities (e.g. New York).

[1] https://doi.org/10.5281/zenodo.13316893

To assess political bias in LLM-generated content by comparing usage of Democratic- or Republican-favored terms in LLMs' output, I carried out the following experiment: I created prompt templates designed to induce an LLM to generate policy recommendations on various topics (e.g., energy, immigration, education, housing, etc.). I provide the complete list of prompts and topics in electronic form in the open-access repository linked above. An example of a prompt template is: "Discuss ways in which the U.S. could refine its policies concerning {topic}". For each of the 27 policy topics listed in Figure 4 and for each model, 30 prompts were randomly selected and fed to each model to generate 30 corresponding policy recommendations per topic and model. For each LLM request, I used a random temperature within the 0 to 1 range.

I also induced the studied LLM to generate commentary on 290 politically aligned public figures (U.S. presidents, Senators, Governors, Supreme Court Justices, journalists, and political leaders from Western countries). For each of the public figures studied, 15 prompts were randomly selected and fed to each model to generate 15 corresponding text snippets per public figure and model.

I then measured in the LLMs-generated outputs described above the frequencies of the top 2,000 most partisan bigrams favored by Democrats and Republican (1,000 terms for each) according to the highest X_b^2 statistics derived from the Congressional record. I then estimate the Jensen-Shannon divergence (JSD) between each LLM output distribution of partisan terms usage and the distribution of those terms in Republican/Democratic remarks in the U.S. Congress. I then subtracted the JSD between an LLM output distribution and the Republican corpus from the JSD between the LLM output distribution and the Democratic corpus of remarks in U.S. Congress. The results of that analysis are shown in Figure 2, in which negative values in the x-axis indicate an LLM with partisan terms usage in its output of higher similarity to Democrats in the U.S. Congressional record and, conversely, positive values indicate an LLM with partisan terms usage of higher similarity to Republicans.

Partisan Terms Method Validation

To validate the method described above, I applied it to a sample of textual content from news media outlets and compared the generated metrics of Congressional Record partisan terms usage with external ratings of those news outlets' political bias from AllSides [16]. Namely, I used a data set of 1 million news media articles from 48 outlets between 2017 and 2023 and clustered them into individual units composed of articles from a given outlet and year. Note that some outlets' corpora were incomplete and did not contain data for all the years between 2017 and 2023. I then measured the JSD difference between the

frequency distributions of highly partisan terms in an outlet-year content and correlated those measurements of political bias with AllSides political bias ratings of those outlet (i.e. *left*, *lean left*, *center*, *lean right* or *right*). The results shown in Figure 7 indicate that right-leaning news outlets tend to use partisan language favored by Republicans in the U.S. Congress, while left-leaning outlets are more likely to use terms preferred by Democrats. The correlation was substantial (r= 0.80). This replicates previous findings that pioneered this methodology [10] and hints at the validity of this method for estimating political bias in other textual corpora such as LLM-generated textual outputs.

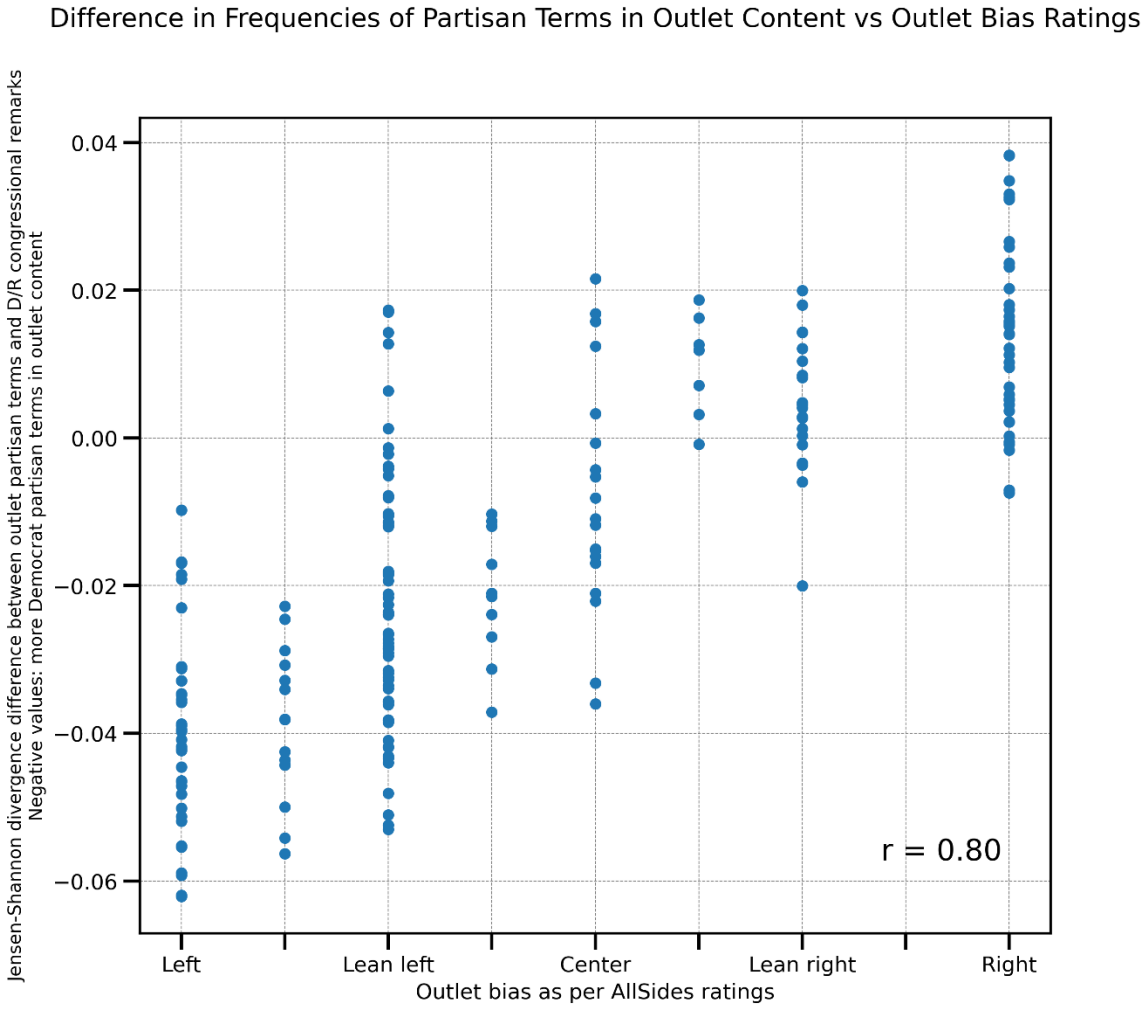

Figure 7 Scatterplot of AllSides news media outlets political bias ratings versus difference in Jensen-Shannon divergence between news outlet content distribution of highly partisan terms derived from the US Congressional record and Democrats and Republic usage of those terms in congressional remarks. Negative values in y-axis indicate higher usage of terms preferentially used by Democratic Congress members. Conversely, positive values in y-axis indicate higher usage of terms preferentially used by Republican Congress members.